International Journal of Medical Informatics 193 (2025) 105681

Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf

Counterfactual reasoning using causal Bayesian networks as a healthcare governance tool

Evangelia Kyrimi [a,*], Somayyeh Mossadegh [b], Jared M. Wohlgemut [b,c], Rebecca S. Stoner [b,c], Nigel R.M. Tai [b,c,d], William Marsh [a]

[a] Department of Electronic Engineering and Computer Science, Queen Mary University of London, United Kingdom
[b] Centre for Trauma Sciences, Blizard Institute, Queen Mary University of London, United Kingdom
[c] Royal London Hospital, Barts Health NHS Trust, London, United Kingdom
[d] Royal Centre for Defence Medicine Birmingham, United Kingdom

ARTICLE INFO

Keywords:
Bayesian Networks
Counterfactual Reasoning
Clinical Decision Support
Healthcare Governance
Quality Assurance

ABSTRACT

Background: Healthcare governance (HG) is a quality assurance processes that aims to maintain and improve clinical practice. Clinical decisions are routinely reviewed after the outcome is known to learn lessons for the future. When the outcome is positive, then practice is praised, but when practice is suboptimal, the area for improvement is highlighted. This process requires counterfactual reasoning, where we predict what would have happened given both what happened and the possible different decisions. Causal models that capture the mechanisms that generate events can support counterfactual reasoning.
Objective: This study is an initial attempt to show how counterfactual reasoning with causal Bayesian networks (CBNs) can be used as a HG tool to assess what would have happened if treatments other than those occurred had been selected.
Methods: Motivated by the Defence Medical Services (DMS) mortality and morbidity (M&M) review meeting, in this paper we (1) extended the use of counterfactual reasoning in CBNs to review decisions, where the alternative treatment strategies and its effect belong to different stages of care, (2) placed counterfactual reasoning in a specific clinical context to examine how it can be used as a HG tool.
Results: Using three realistic examples, we demonstrated how the proposed counterfactual reasoning can be used to assist the DMS M&M review meetings.
Conclusions: Useful lessons can be learned by assessing decisions after they are made. M&M review meetings are fruitful ground for counterfactual reasoning. The use of a clinical decision support tool that can assist clinicians in assessing counterfactual probabilities will be beneficial.

1. Introduction

Counterfactual reasoning regards what would have happened if events other than the ones we are currently observing had happened. Although sometimes claimed otherwise [1], counterfactual reasoning is natural, and it is what makes the human mind unique [2].

In healthcare, counterfactual reasoning is likely to emerge when clinicians experience unexpected or undesirable outcomes [3]. In these circumstances, they may assess what would have happened if a different treatment had been selected. Clinicians may mentally change the perceived antecedents of an outcome and play out the consequences once the facts are known [4].

To be able to ask counterfactual questions models that describe true causal relationships are required [5]. Graphical probabilistic models, such as Bayesian networks (BNs), that model causal relationship, are called Causal Bayesian Networks (CBN) [6]. The contribution of this paper is two-fold; extend the use of counterfactual reasoning in CBNs to review clinical decisions, where the alternative treatment strategies and its effect belong to different stages of care, and demonstrate how counterfactual reasoning with CBNs can be used as a Healthcare Governance (HG) tool to assess what would have happened if treatments other than those occurred had been selected.

* Corresponding author.
E-mail address: e.kyrimi@qmul.ac.uk (E. Kyrimi).
@LinaKyrimi (E. Kyrimi)

https://doi.org/10.1016/j.ijmedinf.2024.105681
Received 20 June 2024; Received in revised form 3 October 2024; Accepted 28 October 2024
Available online 9 November 2024

2. Background

2.1. Counterfactual reasoning with causal Bayesian networks

There are three types of reasoning with BNs: reasoning from observations, interventional [7–9], and counterfactual reasoning. Models that do not have a causally coherent structure cannot support counterfactual reasoning. There are two main approaches for counterfactual reasoning: the pruning theory [5], and the minimal network theory [10]. We believe that when we externally fix a variable to a specific state, it becomes independent of its causes. Thus, only the pruning theory will be considered in this work. Other, less widely used, approaches for performing counterfactual reasoning are described at [11].

The pruning theory distinguishes between exogenous, also known as background variables, and endogenous variables. Background variables are not influenced by other variables in the model but may affect the endogenous variables. Endogenous variables represent the observable factors in the system whose relationships we aim to understand.

Balke and Pearl [12] proposed the use of a twin network, to perform counterfactual reasoning. In a twin network, the actual world is modelled based on observations. In the counterfactual world, we are performing an intervention using the do-operator, do(X = x) in which we manipulate an endogenous variable X by setting it to a specific value x. This action removes all incoming edges to X, effectively isolating it from its influences, which may include both exogenous and endogenous variables. This is known as graph surgery. The transformation can be represented as follows:

$$G_{do(X)} = G \backslash \{\mathrm{Pa(X)} \rightarrow \mathrm{X}\}$$

where Pa(X) includes the set of parent nodes of X.

The two networks have identical structures, except for the arrows towards the endogenous variables that we intervene upon, which are missing in the counterfactual world. Background variables are shared between the two networks, meaning the underlying external factors influencing the system do not change. The background variables help us to connect the twin networks.

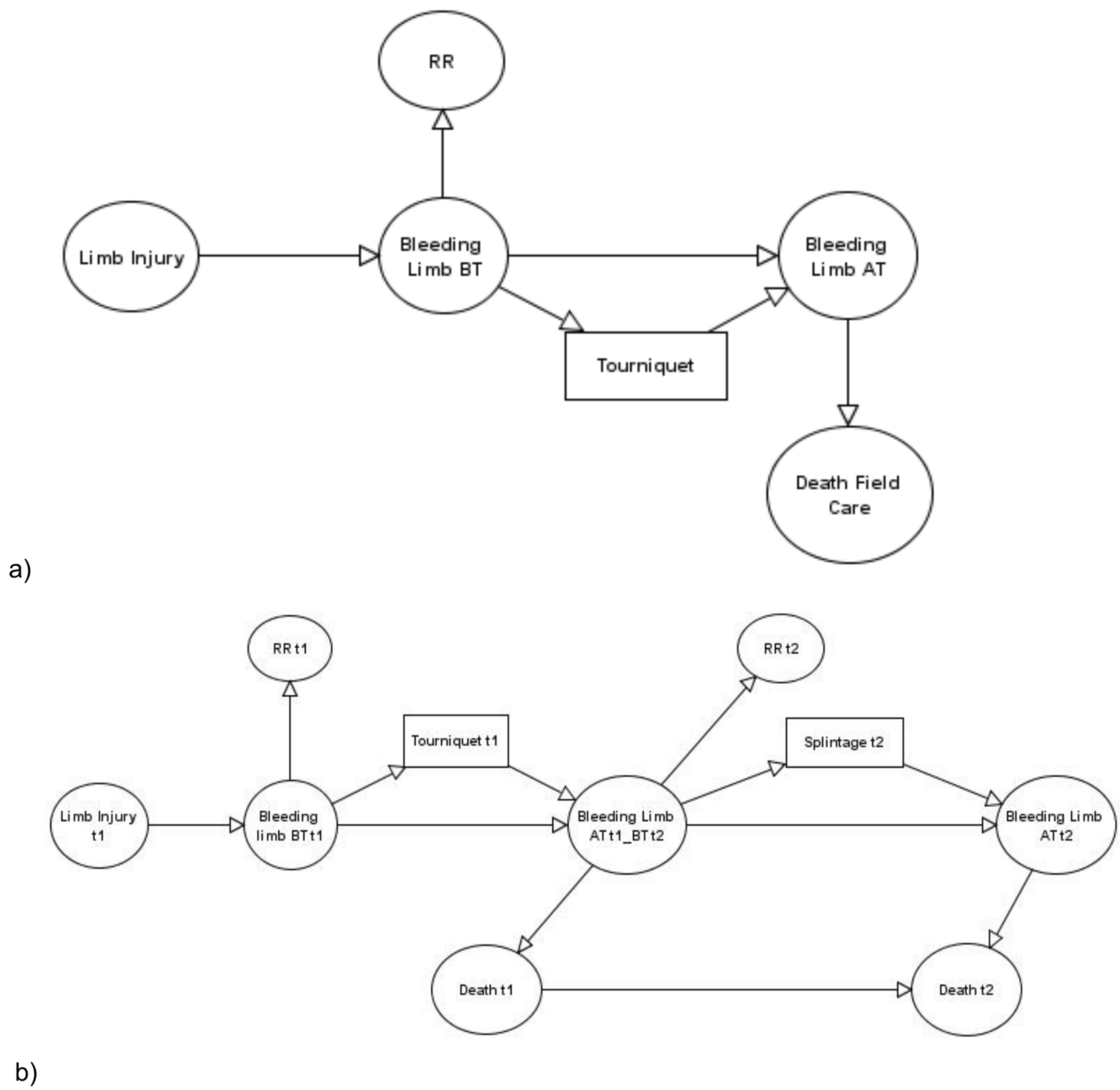

Fig. 1. a. Simplified BN model for predicting the risk of an injured soldier dying during field care. 1b. A BN fragment that captures the likelihood of survival in two successive stages of care.

2.2. Clinical setting: Defence medical services mortality and morbidity review meeting

The aim of the DMS M&M review meeting is to measure the performance of the UK deployed clinical services and to provide beneficial feedback [13]. M&M review meetings are an important healthcare governance component in the British DMS [14] and they are considered as the bed-rock of quality improvement in trauma care [15,16,14].

All operational deaths are reviewed at the M&M review meetings. During these meetings, the review panel asks counterfactual questions to judge the quality of the clinical practice and investigates if anything could have been done better to prevent the undesirable outcome. For instance, questions such as 'If we had performed a thoracostomy, what is the likelihood that the soldier would have survived?' are usually asked [4].

3. Methods

3.1. Counterfactual reasoning with CBNs for reviewing treatment strategies

Let us consider the CBN presented in Fig. 1a. This is a simplified BN that predicts the risk of an injured soldier dying during field care. In the actual world, we observe that although the soldier had a severe limb injury, a tourniquet was not applied, and the soldier died. Imagining a counterfactual world, we would like to know whether the soldier would have survived if we had applied a tourniquet. To answer the counterfactual question, we use a twin network as shown in Fig. 2.

In the counterfactual world the arc going towards the endogenous intervened variable tourniquet is removed. However, the same arc is also removed in the actual world even though this is an observation. If back propagation is maintained then we infer from the non-application of a tourniquet that the injury was not severe; this is not appropriate when we want to model the treatment effect and evaluate its optimality [17]. We therefore reject the point of view that the treatment in the actual world should be simply treated as an observation. Altering the treatment decision cannot affect the variables that have not been caused by the intervention. As a result, all the variables that are an ancestor of the treatment, such as the actual injury, the state of bleeding before treatment (BT) and the related respiratory rate (RR) remain the same in both worlds. These variables are used as background variables.

3.2. Counterfactual reasoning with BNs in successive stages of care

We often might wonder if we had done something different at a specific stage what would have been the outcome at a later stage. Thus, the use of twin networks to perform counterfactual reasoning should be extended for reviewing treatment decisions and their effects in successive stages of patient care. When we review treatment decisions in successive stages of care, the following counterfactual scenarios are possible:

The altered decision and its effect belong to different stages of care
The altered decision and its effect belong to the same stage of care

Suppose that we have the BN in Fig. 1b. This is a simplified BN that predicts the risk of an injured soldier dying in two successive stages of care: field care (t_1) and emergency care (t_2). Let's consider the first scenario. In the actual word, we observe that the soldier had a severe limb injury, a normal RR and even if a tourniquet was not applied in t_1, the soldier was alive at that stage. However, in t_2 his RR became abnormal and although a splint was applied, the soldier died. A

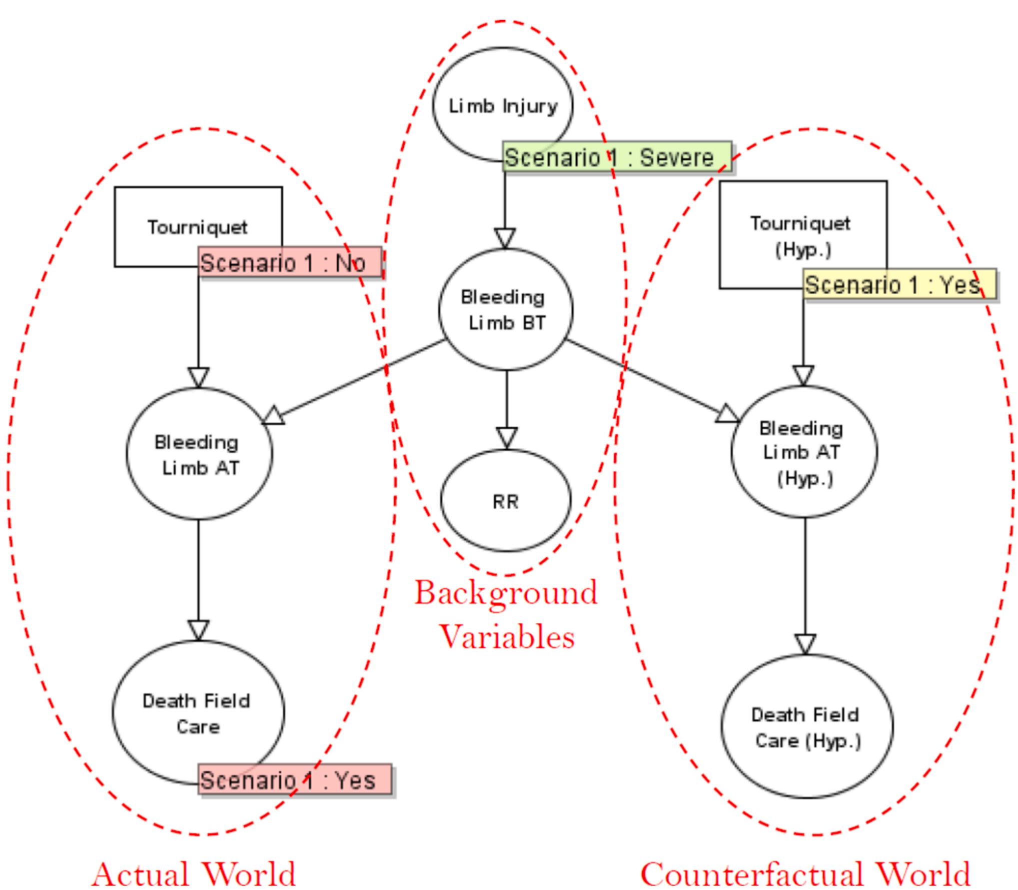

Fig. 2. An illustration of the twin-network method on the BN shown in Fig. 1a.

counterfactual question could be 'What is the likelihood that the soldier would have survived at t_2 if a tourniquet had been applied at t_1?'. We are wondering what would have been the outcome in stage t_2 if we had done something differently in a previous stage.

The twin network that represents this counterfactual question is shown in Fig. 3. One part represents the actual world, and the other part represents the counterfactual world. The variables whose posterior probability stays the same are the background variables. However, extra attention should be paid to all the evidence that follows the altered decision.

In the counterfactual world all the variables that are influenced by the intervened variable, which in our case is tourniquet t_1, should be unobserved. For instance, we cannot assume that RR will stay abnormal in t_2 if a tourniquet was applied at t_1. The same process of reasoning explains why the decision to apply a splintage at t_2 must be also unobserved.

The parents of splintage at t_2 are not the same in the actual and the counterfactual world. In the actual world, this treatment variable is treated as an intervention to represent what happened. However, in the counterfactual world we cannot assume that splintage will have been applied at t_2, so its structure should stay intact. Applying a splintage is influenced by the initial state of bleeding, so the arc that goes from "bleeding" to "splintage" should remain when the treatment variable is unobserved.

In the second scenario, where the altered decision and its effect belong to the same stage of care, two options are available. In case that nothing is observed at a later stage, then a twin network as described at the beginning can be used. If, on the other hand, there is available information at a later stage, then this information should be unobserved in the counterfactual world, following the logic described in the first scenario.

4. Case study

Counterfactual reasoning with CBNs was placed in the context of the DMS M&M review meetings. In this case study, we explain how a CBN can be used to review treatment decisions in successive stages of the soldier's care and help clinicians to answer counterfactual questions. A progressive CBN developed by Kyrimi that predicts soldier's survival in two successive stages of care is going to be used in this case study. The CBN structure is based on expert's knowledge and the use of causal reasoning patterns, which allowed us to have a causally coherent structure. The CBN parameters are learned primarily from data. Expert's knowledge is used when no or not enough data were available. The model's behaviour and performance were evaluated using two different ways: (1) a qualitative way, where the model's behaviour was tested in two extreme scenarios, and (2) a quantitative way, where the average model's accuracy, discrimination and calibration was evaluated. More details can be found at [18]. Our case study is illustrated using three realistic examples:

Example A: review an undesirable outcome in stage t_1 following a possibly sub-optimal clinical decision made in the same stage of care
Example B: review an undesirable outcome in stage t_1 following a reasonable clinical decision made in the same stage
Example C: review an undesirable outcome in stage t_2 following a sub-optimal clinical decision made in the previous stage

The proposed examples are not exhaustive but are sufficient to explain how to answer different counterfactual questions asked during the DMS M&M review meetings.

4.1. Example A

The first example represents a situation where an undesirable outcome happens in field care following a suboptimal clinical practice. Let's assume that there is an explosion in the battlefield and the soldier breaks a long bone (LB) and loses his right leg. Despite the explosion, the soldier has no sign of a head injury; his Glasgow Coma Scale (GCS) is 14 and he has no skull fracture (SF). There was no one near to help him so no tourniquet was applied, and the soldier died. For simplicity, all the other input variables are considered as unobserved. During the M&M review meeting the panel recognises that the soldier died from extensive bleeding, and they wonder "If a tourniquet was applied, what would

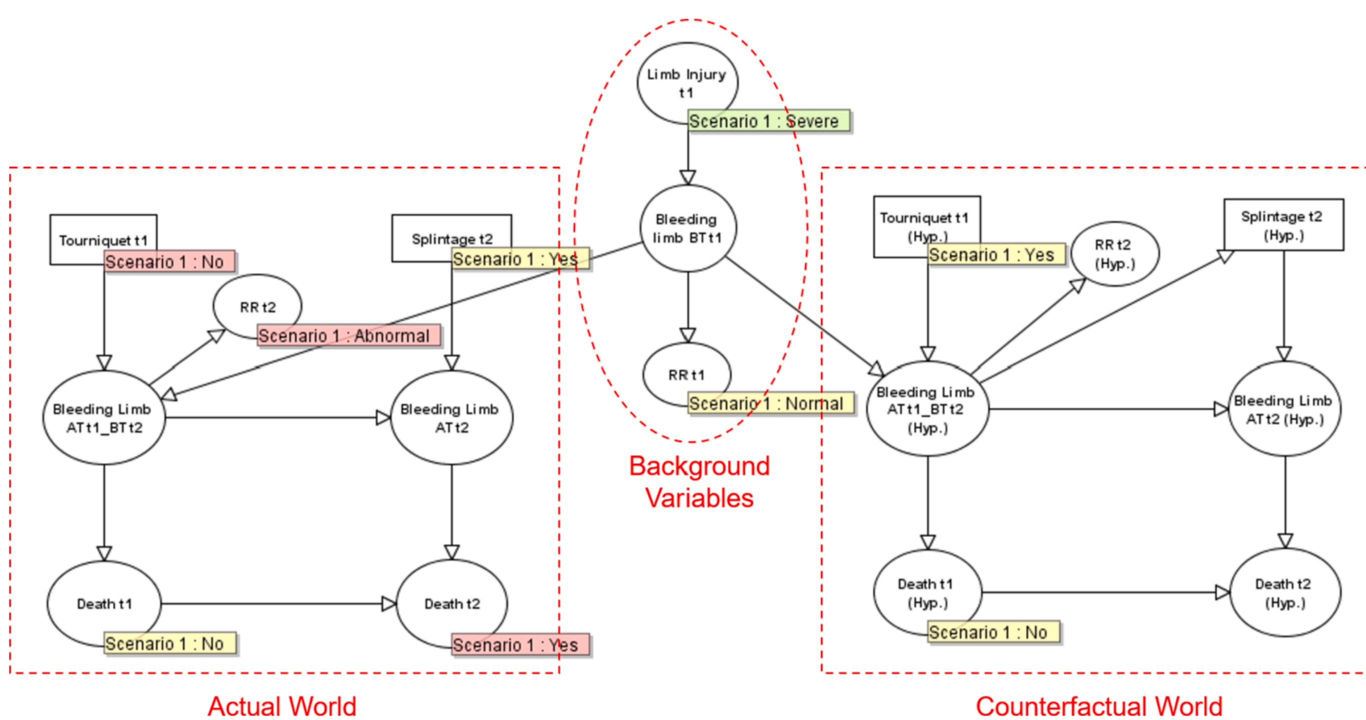

Fig. 3. An illustration of the twin-network method on the BN shown in Fig. 1b.

have been the likelihood of the soldier's survival?". Based on the clinical and medical notes, the review panel would probably have classified the casualty as potentially salvageable. As this scenario takes place in the field, the BN in the first stage is sufficient to answer the counterfactual question. Using a twin network, we illustrate both worlds in Fig. 4. In this query only limb and head injury are considered. The variables that follow the changed decision, applying a tourniquet, are duplicated in both the actual and the counterfactual world. In the counterfactual world, the soldier would probably have survived if a tourniquet had been applied. Using the relative risk, we could say that the soldier in the counterfactual world has 0.14 times the risk of dying compared to a similar situation where a tourniquet was not applied and 2 times the risk of dying compared to the prior, which represents an average situation. Both risks have been calculated as follows:

$$\frac{P(Deadt_1(Hyp.) = Yes|E, do(TourniquetHyp) = Yes)}{P(Deadt_1(Hyp.) = Yes|E, do(TourniquetHyp) = No)} \tag{1}$$

$$\frac{P(Deadt_1(Hyp.) = Yes|E, do(TourniquetHyp) = Yes)}{P(Deadt_1 = Yes)} \tag{2}$$

Not applying a tourniquet was catastrophic. Having the prediction produced by counterfactual reasoning can reassure clinicians that their classification was sensible. In addition, useful lessons can be learned to support better future decisions and clinical guidelines, such as prioritising the use of tourniquets in bleeding limb injuries.

4.2. Example B

The second example is similar to the previous one but now the soldier is also unconscious (GCS = 3) and has a deep skull fracture. No tourniquet was applied, and the soldier died. Here, the review panel wonders again about the effect that a tourniquet would have had on the survival of the soldier. The classification here is not straightforward as, in addition to the bleeding, the soldier has also a severe head injury. The review panel classified the casualty as possibly salvageable. Using the same process as before, we alter the state of tourniquet, and we observe the likelihood of dying. In the counterfactual world, applying a tourniquet reduces the risk of dying because of bleeding, but the soldier is still likely to die because of the head injury. In particular, the soldier in the counterfactual world has 0.22 time the risk of dying compared to a similar situation where a tourniquet was not applied and 13.3 times the risk of dying compared to the prior. Thus, applying the tourniquet may not have been effective in these circumstances because of the head injury.

4.3. Example C

The third example is an extension of the first example. We have the same injury and symptoms in the first stage, a tourniquet was not applied in the first stage, but the soldier survived and was picked up by en route care (second stage). At that point his GCS was worse (GCS = 6) and even though they applied a splintage the soldier died in the helicopter. Now the review panel wonders whether applying a tourniquet at the first stage, would have saved the soldier's life. The twin-network presented in Fig. 5 illustrates both worlds. All the decisions and evidence that happen after the intervened variable are unobserved in the counterfactual world. However, since the comparison focuses on the effect of the alternative decision on the outcome in t_2, we must assume that in the counterfactual world the soldier survived in t_1. In the counterfactual world the soldier has 0.97 times the risk of dying compared to a similar situation where a tourniquet was not applied in the first stage and 0.98 times the risk of dying compared to the prior. Thus, in this alternative scenario, the soldier would have been more likely to have survived in the helicopter if a tourniquet had been applied in the previous stage.

5. Discussion

This work is an initial attempt to explain how counterfactual reasoning with CBNs can be used as a HG tool to assess what would have happened if different treatments had been selected. The novelty of this paper can be summarised as:

- We have extended the use of counterfactual reasoning with CBNs to review clinical decisions, where the alternative treatment strategy and its effect belong to different stages of patient's care.
- We have placed counterfactual reasoning in a specific clinical context, such as the DMS M&M review meetings to potentially be used as a HG tool.

Although we used a military environment, this work can be easily extended to the civilian sector, as mortality and morbidity review meetings are conducted there as well [19].

Despite the benefits of counterfactual reasoning there are some objections. Dawid questioned the scientific validity of counterfactual reasoning and stated that no observable or testable consequences should be regarded as metaphysical [1]. Both Cox and Pearl explained in their commentaries that any counterfactual assumption should not be pressed too far beyond the limits of which they can be tested. In addition, they both claimed that several aspects of counterfactual reasoning can be either directly tested or at least they are indirectly testable via their consequences. Dawid in his rejoinder, although he was still opposed to counterfactual reasoning, agreed with Pearl's argument, and accepted the fact that no problem exists when the counterfactual assumption have testable implications.

In this work, counterfactual reasoning has been used similarly to how clinicians review past decisions. Clinicians may try to remember similar cases that they came across during their careers, where alternative decisions were taken. The observed consequences of those past decisions are used as a verification of their counterfactual reasoning. Except for potentially testable counterfactual events, we focused only on realistic counterfactual questions. As suggested by Dawid, unrealistic counterfactual questions, like 'If the soldier had a normal heart rate, would he still be dead?' were not considered. Finally, we aim for counterfactual reasoning to be used alongside clinicians' judgement and not to replace it. Taking all these factors into consideration, we believe that if the counterfactual questions asked are realistic and ethical, then there is no harm in using counterfactual reasoning with CBNs as an objective HG tool.

The purpose of reflective practice is to learn lessons from past decisions, and improve patient care in the future. Counterfactual reasoning is useful for reflecting on past decisions, as it provides objective evidence of alternate outcomes given differing situations. Counterfactual reasoning with CBNs introduced to M&M review meetings could also improve clinical decision making psychologically. The uncertainty of 'what would have happened if I had done something different' plagues clinicians. It is important to reflect upon past events to prevent future mistakes. But reflection, though important, based on inaccurate counterfactual reasoning, may lead to feelings of guilt (if clinicians believe a better outcome could have occurred) [20,21], or overconfidence (if clinicians believe the best outcome occurred) [22,23]. A CBN that provides accurate outcome prediction utilising counterfactual reasoning, may help objectively assess past decisions.

Despite the relevance of this research there is one main limitation. A more extended evaluation with different CBNs is necessary. In addition, many future directions exist to strengthen further this research. Firstly, using real cases, we should explore how counterfactual reasoning with CBNs can be presented to clinicians and be integrated into the existing workflow. Moreover, it would be beneficial to study the impact that counterfactual reasoning has on various aspects of clinical practice. It would also be useful to investigate how we can perform counterfactual reasoning with CBNs when multiple decisions are reviewed at the same

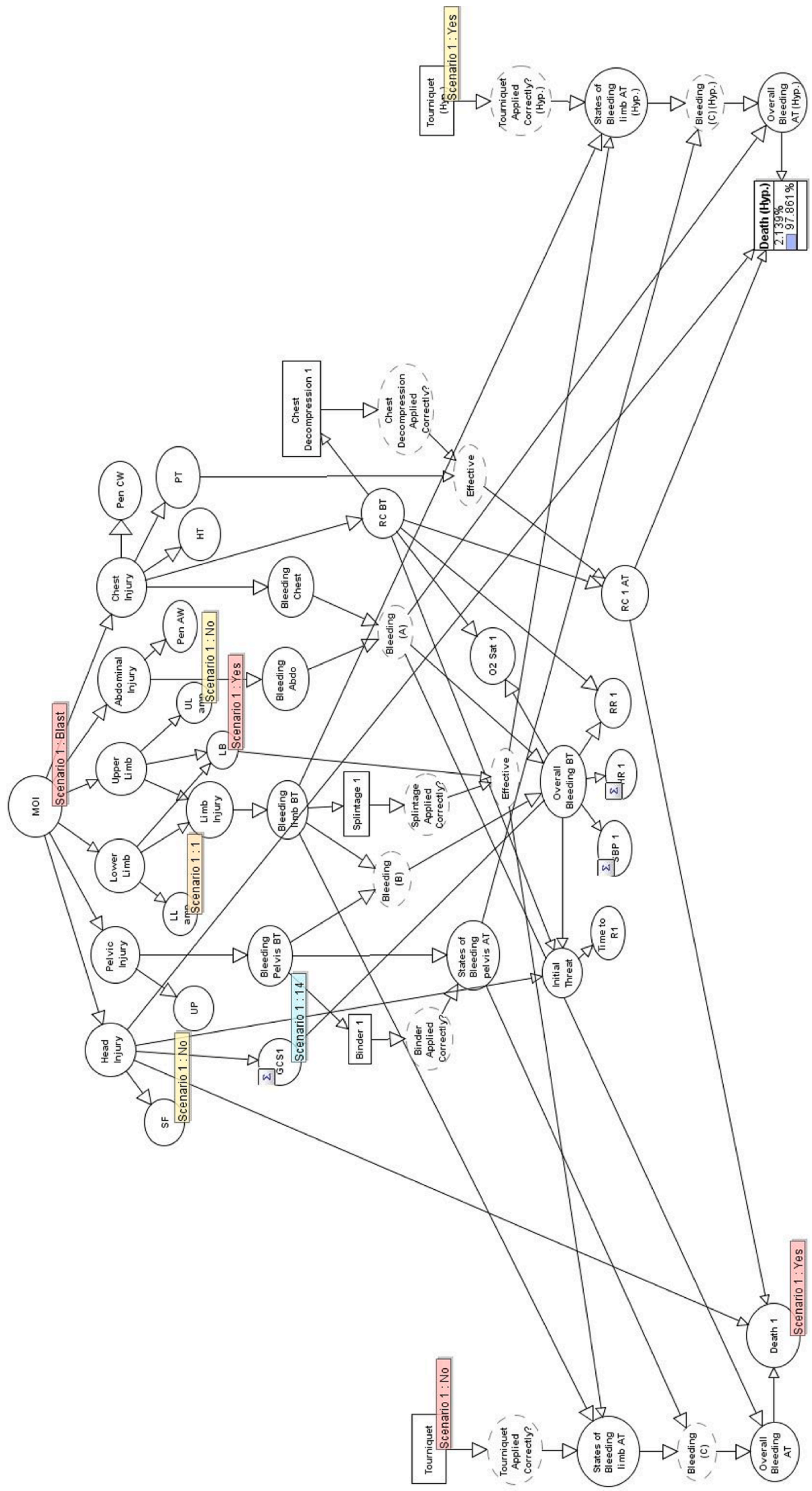

Fig. 4. Twin-network for example A.

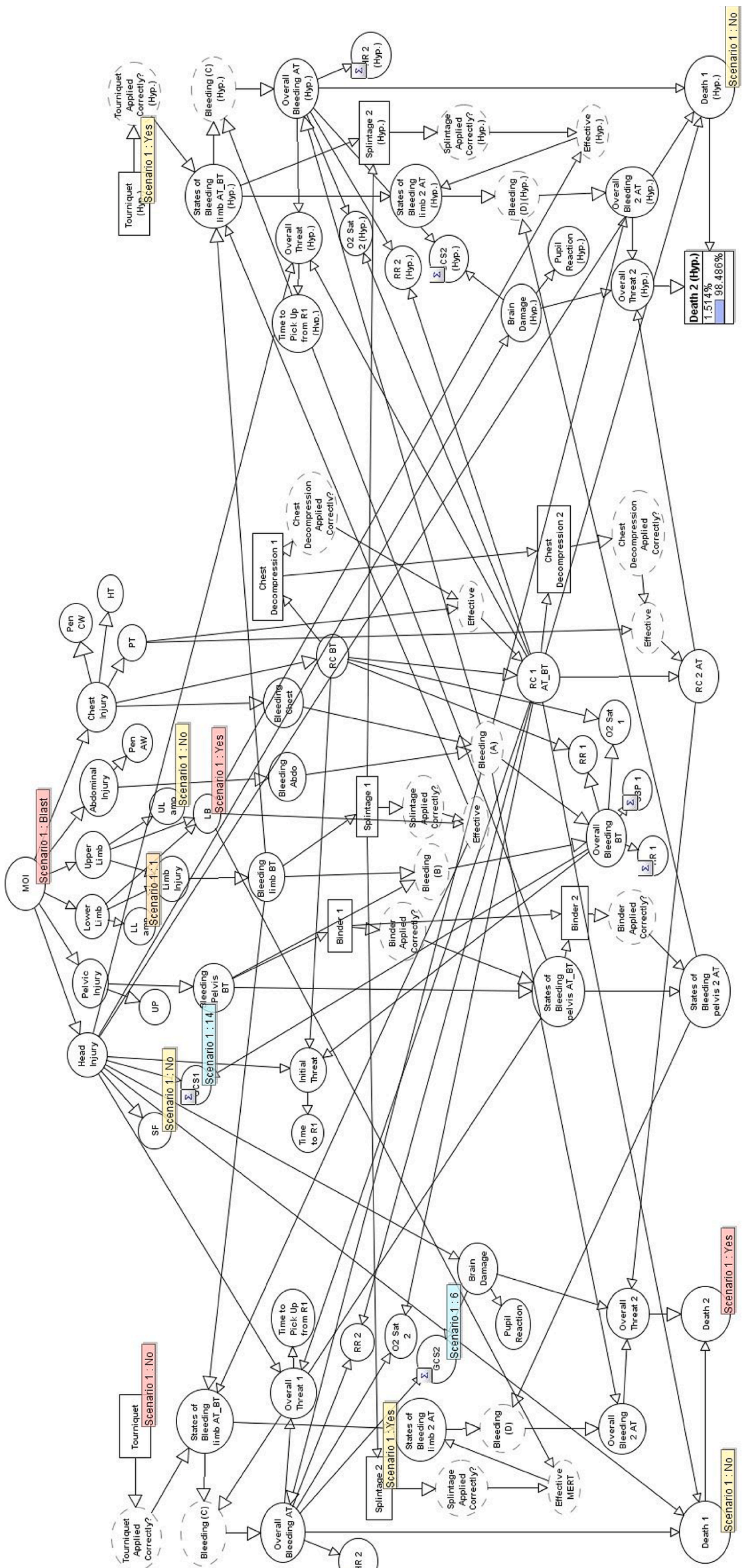

Fig. 5. Twin-network for example C.

time. Finally, we should investigate whether the accuracy of the BN that represents the actual world is enough to reassure an accurate CBN in the counterfactual world.

6. Conclusion

In this paper, we have extended the use of counterfactual reasoning with CBNs to review clinical decisions, where the alternative treatment strategy and its effect belong to different stages of care. We have also placed counterfactual reasoning in a specific clinical context. The use of a tool to assist clinicians in assessing counterfactual probabilities will be beneficial for professional learning and healthcare governance.

Summary Table

What is already known on the topic.	What this study added to our knowledge.
How to perform counterfactual reasoning with causal models using twin networks.	The use of counterfactual reasoning with CBNs to review clinical decisions, where the alternative treatment strategy and its effect belong to different stages of the patient's care.
The necessity of using counterfactual reasoning for answering "what if" questions in healthcare.	Place counterfactual reasoning in a specific clinical context to be used as a healthcare governance tool.

CRediT authorship contribution statement

Evangelia Kyrimi: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Somayyeh Mossadegh:** Writing – review & editing, Resources, Investigation. **Jared M. Wohlgemut:** Writing – review & editing, Resources. **Rebecca S. Stoner:** Writing – review & editing, Resources. **Nigel R.M. Tai:** Writing – review & editing, Supervision. **William Marsh:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

EK, JMW, RSS, NRMT, and WM have received research funding from the United States Department of Defense: a precision trauma care research award from the Combat Casualty Care Research Program of the US Army Medical Research and Materiel Command (W81XWH1920047). The funding agreement ensured the authors' independence in designing the study, interpreting the data, writing, and publishing the report. EK is also funded by Rosetrees Trust and the Royal Academy of Engineering under the Research Fellowship scheme under project ExAIDSS: Explainable AI to ensure trust in clinical Decision Support Systems RF2122-21-226. RSS has received funding from the Royal College of Surgeons of Edinburgh and Orthopaedic Research UK. JMW has received funding from the Royal College of Surgeons of England and Rosetrees Trust. SM acknowledges support from Dept of Research & Clinical Innovation, HQ Joint Medical Grp, UK Defence Medical Services.

References

[1] A.P. Dawid, Causal Inference without counterfactuals, J. Am. Stat. Assoc. 95 (450) (2000) 407–424.
[2] S.A. Sloman, Counterfactuals and causal models: introduction to the special issue, Cogn. Sci. 37 (6) (2013) 969–976, https://doi.org/10.1111/cogs.12064.
[3] S. Sloman, Causal models: How people think about the world and its alternatives, Oxford University Press, USA, 2005.
[4] J.V. Petrocelli, Pitfalls of counterfactual thinking in medical practice: preventing errors by using more functional reference points, J. Public Health Res. 2 (3) (2013), https://doi.org/10.4081/jphr.2013.e24 pp.jphr-2013.
[5] J. Pearl, Causality: Models, Reasoning, and Inference. 2nd edition. 2009.
[6] S.L. Lauritzen, Causal inference from graphical models, Monographs on Stat. Appl. Probability 87 (2001) 63–108, https://doi.org/10.1201/9781420035988.ch2.
[7] P. Spirtes, C. Glymour, and R. Scheines, *Causation, Prediction, and Search*. 1993.
[8] J. Woodward, *Making things happen. A theory of causal explanation.* 2003.
[9] B. Meder, T. Gerstenberg, Y. Hagmayer, M.R. Waldmann, Observing and intervening: rational and heuristic models of causal decision making, Open Psychol. J. 3 (2010) 119–135, https://doi.org/10.2174/1874350101003010119.
[10] E. Hiddleston, A causal theory of counterfactuals, Nous 39 (4) (2005) 632–657, https://doi.org/10.1111/j.0029-4624.2005.00542.x.
[11] C.G. Lucas and C. Kemp, A unified theory of counterfactual reasoning, In: 34th Annual Meeting Cognitive Science Society, 2012, pp. 707–12.
[12] A. Balke and J. Pearl, Probabilistic evaluation of counterfactual queries, In: theTwelfth National Conference on Artificial Intelligence, 1994, pp. 230–237.
[13] R.J. Russell, et al., The role of trauma scoring in developing trauma clinical governance in the Defence Medical Services, Philos. Trans. R. Soc. Lond. B Biol. Sci. 366 (2011) 171–191, https://doi.org/10.1098/rstb.2010.0232.
[14] J. Smith, T. Hodgetts, P. Mahoney, R. Russell, J. Mcleod, Trauma governance in the UK defence medical services, J. R. Army Med. Corps 4 (153) (2007) 239–242.
[15] B.J. Eastridge, et al., Death on the battlefield (2001–2011), J. Trauma. Acute. Care Surg. 73 (6) (2012) S431–S437, https://doi.org/10.1097/TA.0b013e3182755dcc.
[16] R. Russell, N. Hunt, R. Delaney, The mortality peer review panel: a report on the deaths on operations of UK service personnel 2002–2013, J. R. Army Med. Corps 160 (2014) 150–154, https://doi.org/10.1136/jramc-2013-000215.
[17] E. Kyrimi, M.R. Neves, S. McLachlan, M. Neil, W. Marsh, N. Fenton, Medical idioms for clinical Bayesian network development, J. Biomed. Inform. 108 (2020) 103495, https://doi.org/10.1016/j.jbi.2020.103495.
[18] E. Kyrimi, Bayesian Networks for Clinical Decision Making: Support, Assurance, Trust, 2019.
[19] J. Higginson, R. Walters, N. Fulop, Mortality and morbidity meetings: an untapped resource for improving the governance of patient safety? BMJ Qual. Saf. 21 (7) (2012) 576–585, https://doi.org/10.1136/bmjqs-2011-000603.
[20] W.E. Bynum, J.L. Goodie, Shame, guilt, and the medical learner: Ignored connections and why we should care, Med. Educ. 48 (11) (2014) 1045–1054, https://doi.org/10.1111/medu.12521.
[21] R. Sirriyeh, R. Lawton, P. Gardner, G. Armitage, Coping with medical error: a systematic review of papers to assess the effects of involvement in medical errors on healthcare professionals' psychological well-being, Quality Safety Health Care 19 (6) (2010), https://doi.org/10.1136/qshc.2009.035253 e43 e43.
[22] R.P. Vallone, D.W. Griffin, S. Lin, L. Ross, Overconfident prediction of future actions and outcomes by self and others, J. Pers. Soc. Psychol. 58 (4) (1990) 582–592, https://doi.org/10.1037/0022-3514.58.4.582.
[23] M. Brezis, Y. Orkin-Bedolach, D. Fink, A. Kiderman, Does physician's training induce overconfidence that hampers disclosing errors? J. Patient Saf. 15 (4) (2019) 296–298, https://doi.org/10.1097/PTS.0000000000000255.